

Computer Architectures

Exam of 15.9.2025 - part I

First name, Last name, ID.....

Question #1

Let consider the Branch Prediction Mechanism based on the Branch History Table (BHT).

You are requested to

1. Describe the architecture of a BHT
2. Describe the behavior of a BHT, detailing when the processor accesses it, how it selects the right entry, and which information it gets from it
3. Assuming that the processor uses 32 bit addresses, each instruction is 4 byte wide, and the BHT is composed of 8 entries, determine the behavior of the processor if
 - The BHT is initially empty (i.e., full of 0s, meaning NotTaken)
 - The following instructions are executed in sequence (for each instruction the corresponding address is reported)
 - 0x00A60250 l.d r2, v1(r1)
 - 0x00A60254 bnez r2, l1 (branch taken, jumping to 0x00228A00)
 - 0x00228A00 add r2, r2, r4
 - 0x00228A04 bez r2, l2 (branch not taken)
 - 0x00228A08 andi r5, r5, #1
 - 0x00228A0C bez r5, l3 (branch taken, jumping to 0x00469020)
 - 0x00469020 add r1, r2, r3
 - 0x00469024 bez r1, l4 (branch taken, jumping to 0x000092C8)
 - 0x000092C8 add r4, r5, r0
 - 0x000092CC bez r4, l5 (branch taken, jumping to 0x00003A00)
 - 0x00003A00 bez r3, l6 (branch not taken)

You are requested to

- identify the final content of the BHT
- determine which of the branch instructions were correctly predicted, and which were mispredicted.

Initial content of BHT

Row	Content
0	0
1	0
2	0
3	0
4	0
5	0
6	0
7	0

Final content of BHT

Row	Content
0	0
1	1
2	0
3	1
4	0
5	1
6	0
7	0

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Prediction result

Address/Instruction	Accessed BHT line	Prediction (correct/wrong)
0x00582254 bnez r2, 11	5	W
0x00228A04 bez r2, 12	1	C
0x00228A0C bez r5, 13	3	W
0x00469024 bez r1, 14	1	W
0x000092CC bez r4, 15	3	C
0x00003A00 bez r3, 16	0	C